

6. Saturated and Unsaturated Fats

Saturated and unsaturated fats are found in different amounts in different foods.

Unsaturated fats are often described as 'good fats'. When used to replace saturated fats in the diet they help lower cholesterol, one of the risk factors linked to heart disease.

While full-fat dairy products and meat contain large amounts of saturated fats, sources of unsaturated fats include nuts, seeds and vegetable oils.

Saturated and unsaturated fats differ in their chemical structures. Unsaturated fats contain double bonds between the carbon atoms within their molecular structure, whereas saturated fats have no double bonds.



The label on the bottle of a new brand of cooking oil claims that it is better than other cooking oils available on the market because it contains only **15% saturated fat**.

Scientists decided to investigate this claim.

They used bromine water to compare the percentage unsaturation of this new brand of cooking oil with four other vegetable oils, **W**, **X**, **Y** and **Z**. They also tested a known alkane and alkene.

They measured the volume of bromine water that could be added to each oil sample before the bromine water colour remained. Their results are shown below.

Sample	Mean volume of bromine water added (cm ³)	Unsaturation (%)	Saturation (%)
oil W	20.4	50	50
oil X	12.0	30	70
oil Y	24.1	60	40
oil Z	16.3	40	60
new brand oil	30.1	?	?
alkane	0.1	-	-
alkene	40.0	-	-



- (a) Which **one** of these statements best describes the oils tested? Tick (✓) the correct answer. [1]

the oils contain saturated fats only

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the oils contain unsaturated fats only

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the oils contain both saturated and unsaturated fats

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it is not possible to tell whether the oils contain saturated or unsaturated fats

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- (b) Explain why the first drops of bromine water that are added to each of the oil samples are decolourised. [2]

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- (c) The label states that the new oil contains only 15 % saturated fat. Use the results to show whether this statement is correct. [3]

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- (d) Explain why these results might persuade many consumers to use this new brand of oil in preference to the other available brands. [2]

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